PROJECT REPORT

Exploratory Data Analysis of Colombo Stock Exchange (CSE) Data

ITDS22013 - Data Analysis with Python

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1. Introduction

The Colombo Stock Exchange (CSE) plays an important role in Sri Lanka's financial market by enabling the trading of listed securities and reflecting changes in investor activity and market conditions. This report presents a data analysis of historical CSE market indicators using Python. The analysis focuses on the All-Share Price Index (ASPI), S&P SL20 Index, daily turnover, volume of turnover, number of transactions, market capitalization, and market price earnings ratio.

The purpose of the analysis is to clean and prepare the historical dataset, examine the behavior of key market variables, identify trends and relationships within the data, and obtain meaningful insights through statistical and visual analysis. Python libraries including Pandas, NumPy, Matplotlib and Seaborn were used throughout the analysis.

2. Objectives of the Data Analysis

- To import and prepare historical Colombo Stock Exchange data for analysis.
- To identify and handle missing or inconsistent values in the dataset.
- To calculate descriptive statistics for major stock market indicators.
- To analyze changes in the All-Share Price Index overtime.
- To investigate relationships between different market indicators.
- To identify patterns, trends and correlations using appropriate visualizations.
- To interpret the results and obtain meaningful insights about CSE market behavior.

3. Data Description

3.1 Data Source

Source: Central Bank of Sri Lanka (CBSL) Economic Data Library (Financial Sector - Share Market Data)

These were share market historical data for Colombo Stock Exchange The initial data was in the Data.xls file, with market data across various dates. While performing the pre-processing steps, data must restructure such that each row consists of trading day and column will be the market indicator.

3.2 Dataset Size

Number of observations: 1,379
Time period: 2 January 2019 – 18 December 2024
Number of analyzed variables: 7

3.3 Variables

Variable	Description
ASPI	All Share Price Index representing the overall CSE market
S&P SL20	Index representing selected large and liquid companies
Turnover	Daily value of securities traded
Volume	Number of securities traded during the day
Trades	Number of transactions completed
Market Capitalization	Overall market value of listed securities
PE Ratio	Market Price Earnings Ratio

Figure 1: Sample of the cleaned CSE dataset

Item Name	EQUITY- All share price index	EQUITY-S&P SL20 Index	EQUITY- Daily Turnover	EQUITY- Volume of Turnover (No.)	EQUITY- No. of Transactions (Trades)	EQUITY-Market Capitalization	EQUITY- Market Price Earnings Ratio
2019-01-02	6062.20	3111.07	211.50	12061968.0	4110.0	2844.06	9.7
2019-01-03	6058.48	3091.52	804.67	12890699.0	3167.0	2842.31	9.7
2019-01-04	6067.66	3105.75	68.78	3145002.0	3155.0	2846.62	9.7
2019-01-07	6022.99	3070.20	405.07	7367386.0	3320.0	2825.67	9.6
2019-01-08	5992.36	3049.95	932.96	18393744.0	3227.0	2811.29	9.6

4. Data Preprocessing and Cleaning

4.1 Removal of Unnecessary Columns and Rows

The initial data set had redundant columns (Unnamed: 0, Unit, and Scale) which were not required for this analysis, hence they were dropped. Rows where item names were missing and rows indicating the 'category' as "Share Market Data-Overall Data" were removed.

4.2 Data Restructuring

In the original dataset, market indicators were arranged as rows while dates appeared as columns. The dataset was therefore transposed so that each trading date became an individual observation and each market indicator became a feature. The date values were then converted to Python datetime format.

4.3 Renaming Variables

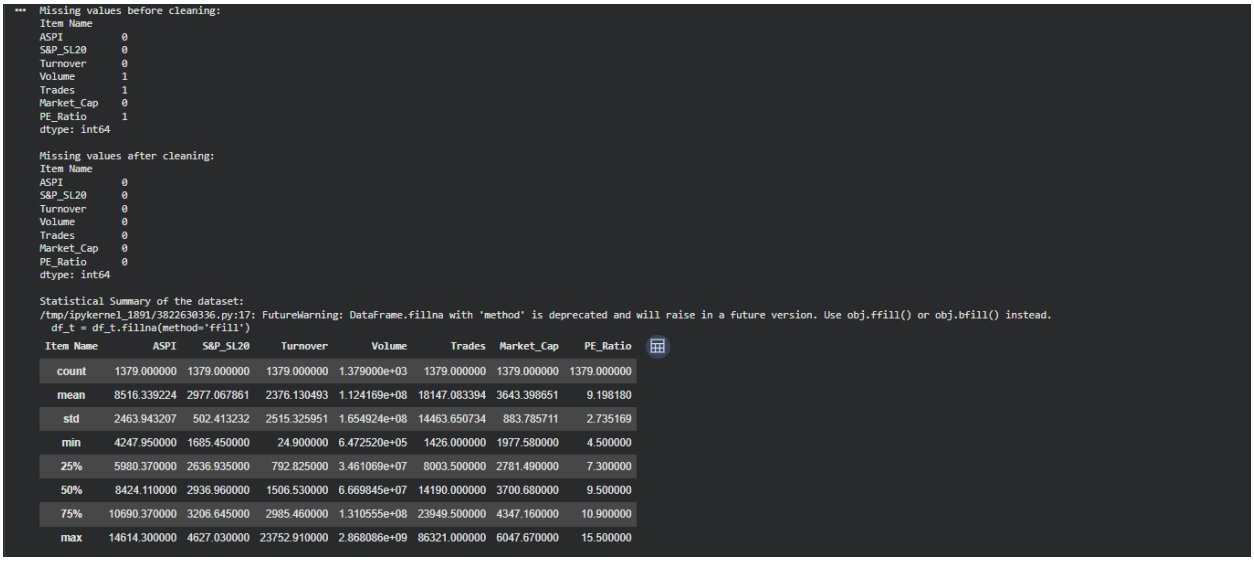
EQUITY- All share price index → ASPI
EQUITY-S&P SL20 Index → S&P_SL20

4.4 Missing Value Handling

Missing-value analysis identified one missing observation each in Volume, Trades and PE Ratio. Since the dataset represents chronological market observations, forward filling was applied, where the most recently available value was used to replace the missing observation. Following this process, no missing values remained in the analysed dataset.

- ASPI – 0
- S&P SL20 – 0
- Turnover – 0
- Volume – **1**
- Trades – **1**
- Market Capitalization – 0
- PE Ratio – **1**

Figure 2: Missing values before and after data cleaning



```
Missing values before cleaning:
Item Name
ASPI          0
S&P_SL20      0
Turnover      0
Volume        1
Trades        1
Market_Cap    0
PE_Ratio      1
dtype: int64

Missing values after cleaning:
Item Name
ASPI          0
S&P_SL20      0
Turnover      0
Volume        0
Trades        0
Market_Cap    0
PE_Ratio      0
dtype: int64

Statistical Summary of the dataset:
/tmp/ipykernel_1891/3822630336.py:17: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a future version. Use obj.ffill() or obj.bfill() instead.
  df_t = df_t.fillna(method='ffill')
```

Item Name	ASPI	S&P_SL20	Turnover	Volume	Trades	Market_Cap	PE_Ratio
count	1379.000000	1379.000000	1379.000000	1.379000e+03	1379.000000	1379.000000	1379.000000
mean	8516.339224	2977.067861	2376.130493	1.124169e+08	18147.083394	3643.398651	9.198180
std	2463.943207	502.413232	2515.325951	1.654924e+08	14463.650734	883.785711	2.735169
min	4247.950000	1685.450000	24.900000	6.472520e+05	1426.000000	1977.580000	4.500000
25%	5980.370000	2636.935000	792.825000	3.461069e+07	8003.500000	2781.490000	7.300000
50%	8424.110000	2936.960000	1506.530000	6.669845e+07	14190.000000	3700.680000	9.500000
75%	10690.370000	3206.645000	2985.460000	1.310555e+08	23949.500000	4347.160000	10.900000
max	14614.300000	4627.030000	23752.910000	2.868086e+09	86321.000000	6047.670000	15.500000

5. Descriptive Statistical Analysis

Indicator	Mean	Minimum	Maximum
ASPI	8,516.34	4,247.95	14,614.30
S&P SL20	2,977.07	1,685.45	4,627.03
Daily Turnover	2,376.13	24.90	23,752.91
Trades	18,148.47	1,426	86,321
Market Capitalization	3,643.40	1,977.58	6,047.67
PE Ratio	9.20	4.50	15.50

Descriptive Analysis Summary shows that the CSE marketactivity fluctuated broadly between 2019 & 2024. The average value of the ASPI was roughly 8,516.34, ranging from 4,247.95 to 14,614.30, which is ample that our CSE Market showed dynamic movement during the examined years. Similarly, the average Daily Turnover exceeded LKR2,376.13 million up to LKR23,752million which indicate extremely busy days that appeared in CSE Market. The daily Number of transactions has also fluctuated in this periods which goes from 1,426 to 86,321 transactions per day.

Table 2: Descriptive statistics of the CSE market indicators

Item Name	ASPI	S&P_SL20	Turnover	Volume	Trades	Market_Cap	PE_Ratio
count	1379.000000	1379.000000	1379.000000	1.379000e+03	1379.000000	1379.000000	1379.000000
mean	8516.339224	2977.067861	2376.130493	1.124169e+08	18147.083394	3643.398651	9.198180
std	2463.943207	502.413232	2515.325951	1.654924e+08	14463.650734	883.785711	2.735169
min	4247.950000	1685.450000	24.900000	6.472520e+05	1426.000000	1977.580000	4.500000
25%	5980.370000	2636.935000	792.825000	3.461069e+07	8003.500000	2781.490000	7.300000
50%	8424.110000	2936.960000	1506.530000	6.669845e+07	14190.000000	3700.680000	9.500000
75%	10690.370000	3206.645000	2985.460000	1.310555e+08	23949.500000	4347.160000	10.900000
max	14614.300000	4627.030000	23752.910000	2.868086e+09	86321.000000	6047.670000	15.500000

6. Exploratory Data Analysis

6.1 ASPI Trend Over Time

Figure 3: All Share Price Index (ASPI) over time, 2019–2024

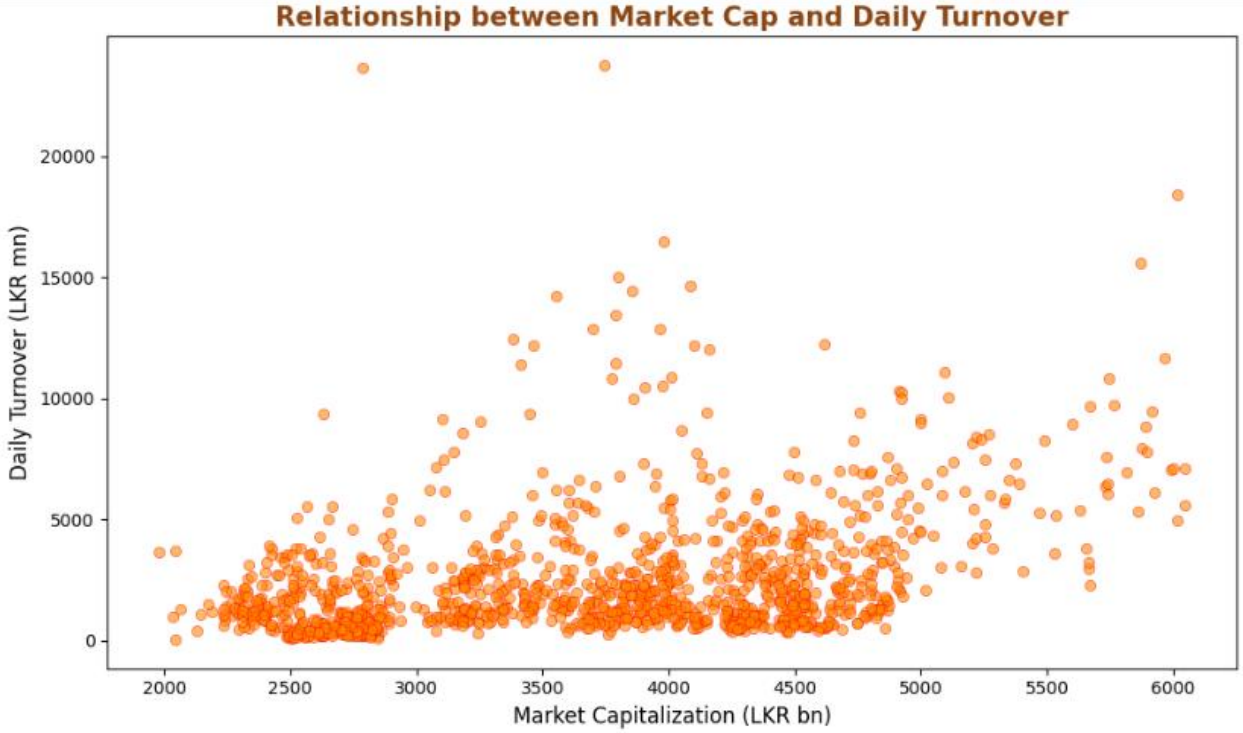


The movement of ASPI during the period of study is given in figure 3. ASPI was not constant over time; there was a good proportion of fall and then raise in ASPI values at different periods. In overall the trend indicates the Sri Lanka equity market has been in changing circumstances and at towards the end of the data-set there are higher rise and has reach the highest values. From our dataset minimum of ASPI was 4,247.95 and highest value of 14,614.30 over the period of 6 years has a highest variation which shows an important feature over six year period for an individual over a term of 6 year time period, which could have been considered as long-term

variation within a market that is dynamic in nature and exhibits a good degree of short-term and moderate-term variations.

6.2 Relationship Between Market Capitalization and Daily Turnover

Figure 4: Relationship between Market Capitalization and Daily Turnover

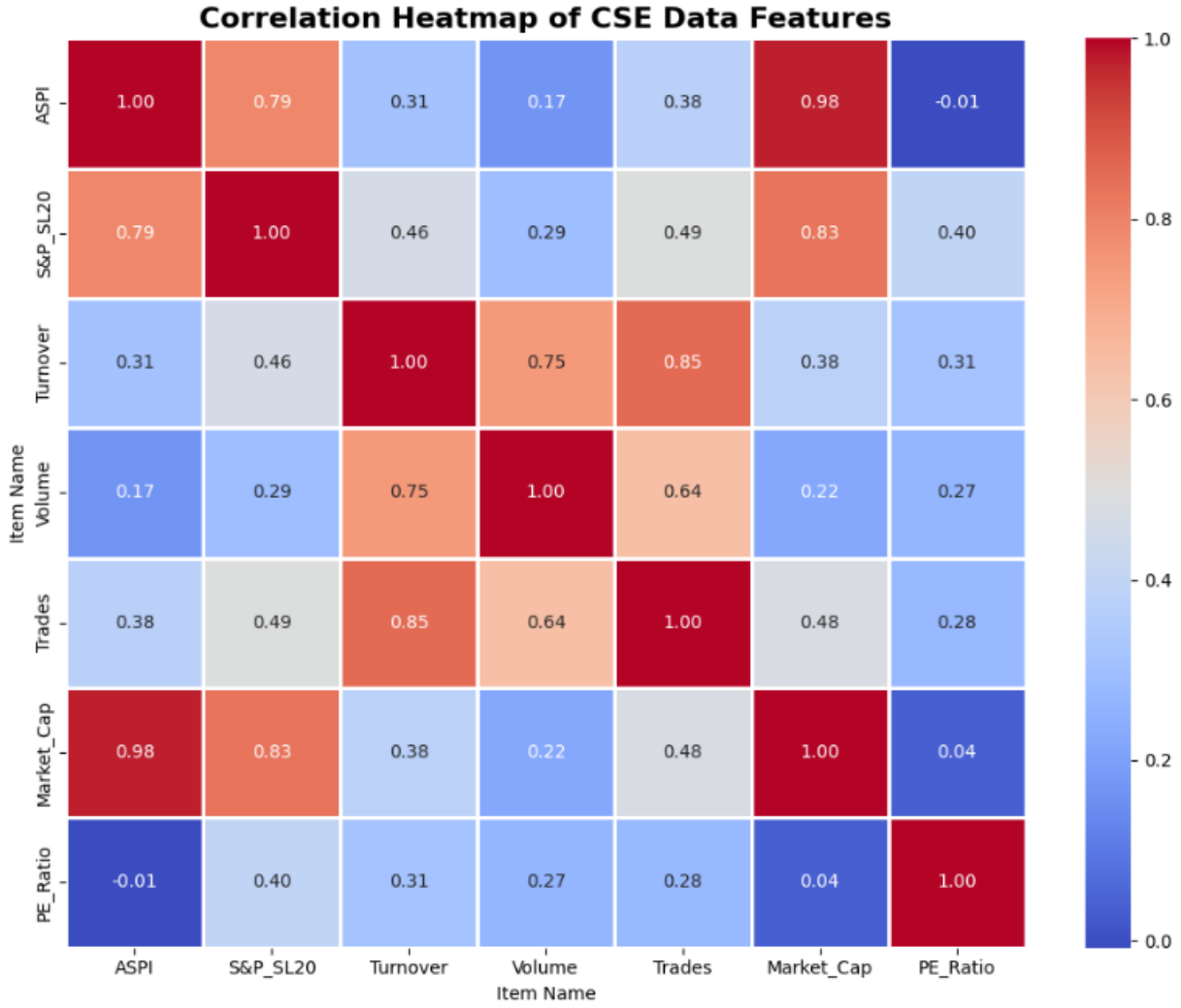


The scatter plot was used to investigate whether higher market capitalization was associated with greater daily trading turnover. The distribution of points indicates that turnover can vary substantially even at similar levels of market capitalization. Several unusually high turnover observations can also be identified, suggesting that individual trading days may experience exceptionally high levels of market activity.

Therefore, market capitalization alone does not appear to explain all changes in daily turnover. Other market conditions and investor behavior may also contribute to changes in trading activity.

6.3 Correlation Analysis

Figure 5: Correlation heatmap of CSE market indicators



Pearson correlation analysis was conducted to examine the strength and direction of the linear relationships among the major Colombo Stock Exchange market indicator. The correlation coefficients range between -1.0 to +1.0 where higher values nearer +1 represents a stronger positive association between variables, values closer to -1 represent a stronger negative association and values nearing 0 represents a low and linear association between variables. In terms of interpretation of correlation coefficients Values from +0.7 to +1 strongly positive correlation Values from +0.3 to +0.6 positive correlation Values from -0.3 to +0.3 nearly zero linear correlation Value from -0.6 to -0.3 negative correlation Value from -0.7 to -1.0 strongly negative correlation. The correlations were calculated using the daily market observations available from 2 January 2019 to 18 December 2024 show that ASPI and market capitalization are positively correlated with r=0.98.

This indicate a very strong positive relation so the higher values of ASPI associated higher values of total market capitalization during period analysed..

There is a strong positive relationship between ASPI and S&P SL20 Index with r=0.79 indicating the higher movement of the broad market index correlate with movement of larger listed stocks during the periods. Between the daily turnover and the number of trades the correlation is a strong positive one (r=0.85) which shows an increase in total market value is associated by increase in total number of transactions executed in the market on any given day.. Turnover and volume also show relatively high relation (r=0.75) and positive relation with number of trades was evident between volume and number of trades (r=0.64). Between the S&P SL20 and Market Capitalization there was a strong positive correlation (r=0.83).

Meanwhile Price Earnings Ratio do not show strong linear relation with ASPI (r=-0.01).

Also Price Earnings Ratio and market capitilisation have weak positive correlation (r=0.04). Over all in the selected period market indicators have shown that ASPI and S&P SL20 and Market capitalisation are strongly linearly connected to each other. Variables related to turnover, volume and number of trades also seem to have some relation among each other.

The correlations are just indicative for relationships but not indicative of a causality effect between variables.

Relationship	**Correlation**	**Interpretation**
ASPI ↔ Market Capitalization	**0.98**	Very strong positive
ASPI ↔ S&P SL20	**0.79**	Strong positive
S&P SL20 ↔ Market Capitalization	**0.83**	Strong positive
Turnover ↔ Trades	**0.85**	Strong positive
Turnover ↔ Volume	**0.75**	Strong positive
Volume ↔ Trades	**0.64**	Moderate/strong positive
ASPI ↔ Trades	0.38	Weak/moderate positive
ASPI ↔ Turnover	0.31	Weak positive
ASPI ↔ Volume	0.17	Weak positive
ASPI ↔ PE Ratio	**-0.01**	Almost no relationship

7. Key Findings and Discussion

The preliminary study identified the following trends of the Colombo Stock Exchange dataset. First, the values of the ASPI were observed to be highly volatile within the timespan 2019 to 2024, implying that there were fluctuations in the market over the time. The values of the index fluctuated between the values of 4248 and 14614 approximately.

Secondly, trading volume also fluctuated in the period analyzed. Both the number of transactions and the turnover in the stock market were seen to have large variations and relatively large standard deviations. This can imply that certain normal trading days were affected by periods of abnormal market activity.

Thirdly, the market capitalization also varied significantly during this time span of 2019 to 2024. It was seen to vary between 1977.58 and 6047.67. The scatter plot also suggested that there could be the same market capitalization values corresponding to different trading volume values, which implies market activity depends on several factors.

Last but not the least, the correlation analysis summarized the correlation between the values of the ASPI, S&P SL20, turnover, volume, trades, market capitalization and the PE ratio.

8. Conclusion

This report analyzed historical Colombo Stock Exchange data using Python-based data analysis techniques. The dataset was successfully cleaned, restructured and prepared before conducting descriptive and exploratory analysis. Missing observations were identified and handled, while statistical summaries were used to understand the distribution and variability of the main market indicators.

Visual analysis demonstrated significant changes in the ASPI over time as well as variations in daily trading activity, market capitalization and other market indicators. Scatter plot and correlation analyses also provided insight into the relationships among the variables. Overall, the analysis demonstrates that CSE market behavior varies considerably over time and that multiple indicators are useful when examining market performance.

The cleaned and analyzed dataset provides a reliable basis for the separate machine learning component of the assignment, where predictive models can be developed to estimate future market behavior.

9. References

[1] Central Bank of Sri Lanka (CBSL), *Economic Data Library: Financial Sector Share Market Data (CSE Historical Datasets)*. Colombo, Sri Lanka, 2024.

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[3] W. McKinney, "Data Structures for Statistical Computing in Python," in *Proceedings of the 9th Python in Science Conference*, 2010, pp. 51–56.

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[5] J. D. Hunter, "Matplotlib: A 2D Graphics Environment for Data Visualization," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

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10. Appendix

Appendix A – Selected Python Code
Data Loading and Preprocessing Code snippet

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Read the file as an HTML table (because the CBSL .xls file is actually in HTML format)
tables = pd.read_html('Data.xls')
df = tables[1] # The actual data is in the second table

# Remove unnecessary columns to make the dataset clean
cols_to_drop = [c for c in ['Unnamed: 0', 'Unit', 'Scale'] if c in df.columns]
df = df.drop(columns=cols_to_drop)

# Remove rows that do not have an 'Item Name' or contain category headers
df = df.dropna(subset=['Item Name'])
df = df[df['Item Name'] != 'Share Market Data-Overall Data']

# Set the 'Item Name' as the index and transpose (swap rows and columns)
# so that Dates become rows and Features become columns
df.set_index('Item Name', inplace=True)
df_t = df.T

# Convert the row index into proper Date format
df_t.index = pd.to_datetime(df_t.index, errors='coerce')
df_t = df_t.dropna(how='all') # Drop any rows where the date is missing

# Convert all values to numbers (floats) for calculations
df_t = df_t.apply(pd.to_numeric)

# Display the first 5 rows to check the data
df_t.head()
```